

Supplemental Material for “Cold Crystallization of the Lithium Sublattice and the Warming-Only Resistance Anomaly of Delithiated Li_xCoO_2 ”

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This document gives the derivations that the main text quotes, in the order the main text uses them. Equations are numbered S1 onward and are referred to from the main text by those numbers. Section S7 lists every symbol once, with its dimension, so that no symbol carries two meanings across the two documents.

S1. MIXTURE FORMULA AND THE TWO-PROBE CIRCUIT

A. Bounds on a two-phase channel

Let the channel have length L and cross-sectional area A , and let a fraction ϕ of its volume have transformed from the parent phase P, of resistivity ρ_P , into the ordered phase O, of resistivity ρ_O . Two microgeometries bound every other one [1, 2]. In the series arrangement the transformed material forms slabs that span the cross section, so the resistances add,

$$R_{\text{series}} = \frac{L}{A} [(1 - \phi)\rho_P + \phi\rho_O]. \quad (\text{S1})$$

With $R_P \equiv \rho_P L/A$ the resistance the same channel would have with nothing transformed, and with the resistivity ratio

$$s \equiv \frac{\rho_O}{\rho_P}, \quad (\text{S2})$$

Eq. (S1) gives the excess exactly,

$$\frac{\Delta R}{R_P} \equiv \frac{R_{\text{series}} - R_P}{R_P} = \phi(s - 1). \quad (\text{S3})$$

In the parallel arrangement the transformed material forms filaments running the length of the channel, so the conductances add, and

$$\frac{\Delta R}{R_P} = \frac{\phi(1 - 1/s)}{1 - \phi(1 - 1/s)} \xrightarrow{\phi(1-1/s) \ll 1} \frac{\phi(s - 1)}{s}. \quad (\text{S4})$$

Both bounds have the form quoted as Eq. (2) of the main text, $\Delta R/R_P = c\phi(s - 1)$, with the geometry factor $c = 1$ for the series limit and $c = 1/s$ for the parallel limit, so that $c \in (0, 1]$ whenever the transformed phase is the more resistive one. Both expressions vanish at $\phi = 0$ and at $s = 1$, as they must, and both are monotonic in ϕ and in s . The quantity the experiment supplies is the product $c\phi(s - 1)$, which the zero-field maximum places at approximately unity; the individual factors are not separately determined by a single trace.

B. What the two-probe geometry allows

The measured resistance is the sum in Eq. (1) of the main text, $R_{\text{meas}} = R_c + R_{\text{ch}}$, where R_c collects the two Schottky junctions evaluated on the quasilinear branch of their current-voltage characteristic and R_{ch} is the channel. Writing the warming and cooling traces separately and subtracting,

$$\Delta R = [R_c^w - R_c^c] + [R_{\text{ch}}^w - R_{\text{ch}}^c], \quad (\text{S5})$$

so that any contact resistance that depends on temperature but not on thermal history cancels identically. What does not cancel is a contact whose barrier height itself depends on the lithium arrangement beneath the electrode, since that arrangement carries the same history dependence as the channel. Equation (S5) is therefore an upper bound on the channel contribution, and every quantitative statement in the main text is made about ΔR rather than about a resistivity. The two experiments that separate the terms are channel-length scaling, under which the second bracket grows linearly with L while the first does not, and a four-probe measurement, which removes the first bracket outright.

S2. UNFREEZING ON THE WARMING BRANCH

A. Growth from a frozen nucleus population

Start from the growth velocity of Eq. (7) of the main text, with the sample quenched into the untransformed state and the ordered phase thermodynamically favored below T_{ord} :

$$v(T) = v_0 e^{-E_a/k_B T} q(T), \quad q(T) \equiv 1 - e^{\Delta f/k_B T}, \quad (\text{S6})$$

in which $v_0 = \nu a$ is the velocity a domain boundary would have if every attempt succeeded, $a = 0.28$ nm being the lithium-lithium spacing in the plane, and $\Delta f = f_O - f_P$ is the free-energy difference per cobalt, negative below T_{ord} . The mobility factor counts the attempts that clear the migration barrier and the thermodynamic factor q removes the reverse hops, so that v vanishes at T_{ord} as it must and approaches $v_0 e^{-E_a/k_B T}$ only at undercoolings large enough that $|\Delta f|$ exceeds $k_B T$. Close to the boundary $|\Delta f| \simeq \Delta S(T_{\text{ord}} - T)$, so that $q = 1 - e^{-\sigma(T_{\text{ord}} - T)/T}$ with $\sigma = \Delta S/k_B$ per cobalt.

Nucleation is complete before the warming sweep begins, since the cooldown delivers the sample to base temperature already seeded, so the transformation is site saturated: each of the n_0 nuclei per unit area grows as a disk of the same radius,

and the Kolmogorov–Johnson–Mehl–Avrami construction of Sec. S3 collapses to

$$\phi(T) = 1 - \exp[-\pi n_0 R_w(T)^2], \quad (\text{S7})$$

with R_w the radius accumulated since the start of the sweep. On a linear sweep $T(t) = T_0 + rt$ with $r > 0$ the substitution $dt' = dT'/r$ turns that radius into a temperature integral,

$$R_w(T) = \frac{1}{r} \int_{T_0}^T v(T') dT' \equiv a \Theta(T), \quad (\text{S8})$$

$$\Theta(T) = \frac{v}{r} \int_{T_0}^T e^{-E_a/k_B T'} q(T') dT'.$$

The integrand of Θ has dimensions of inverse time, the integration variable has dimensions of temperature, and r has dimensions of temperature per time, so Θ is dimensionless, as it must be if R_w is to carry the dimensions of a . Its physical meaning is the number of successful rearrangement attempts made between T_0 and T , net of the reverse hops that q subtracts, and $R_w = a\Theta$ is the distance a domain boundary advances over the same interval.

B. Asymptotic evaluation

Change variable to $y = E_a/k_B T'$, so that $T' = E_a/k_B y$ and $dT' = -(E_a/k_B) y^{-2} dy$. Equation (S8) becomes

$$\Theta(T) = \frac{v E_a}{r k_B} \int_{y_0}^y q\left(\frac{E_a}{k_B y}\right) \frac{e^{-y'}}{y'^2} dy', \quad y = \frac{E_a}{k_B T}, \quad y_0 = \frac{E_a}{k_B T_0}. \quad (\text{S9})$$

Because $y_0 > y$ and the integrand falls exponentially, the lower limit of the sweep contributes a fraction $e^{-(y_0-y)}(y/y_0)^2 q(T_0)/q(T)$ of the total. For the parameters of interest, $E_a = 0.387$ eV, $T = 146$ K and $T_0 = 80$ K, that fraction is 7.5×10^{-12} , so the lower limit may be replaced by infinity. The thermodynamic factor varies slowly on the scale over which the exponential does, and is carried to first order by writing the integrand as e^h with $h = \ln q - E_a/k_B T'$; the pure exponential part has the standard asymptotic expansion

$$\int_y^\infty \frac{e^{-y'}}{y'^2} dy' = \frac{e^{-y}}{y^2} \left[1 - \frac{2}{y} + O(y^{-2}) \right], \quad (\text{S10})$$

obtained by integrating by parts once and bounding the remainder. Substituting Eq. (S10) into Eq. (S9), restoring $y = E_a/k_B T$ and keeping the first derivative of q gives Eq. (4) of the main text together with both of its leading corrections,

$$\Theta(T) = \frac{v}{r} \frac{k_B T^2}{E_a} q(T) e^{-E_a/k_B T} \left[1 - \frac{2k_B T}{E_a} - \frac{k_B T^2}{E_a} \frac{d \ln q}{dT} + \dots \right] \quad (\text{S11})$$

The two corrections are of the same size and of opposite sign, and neither may be discarded on the ground that the other is small. At $T = 146$ K and $E_a = 0.387$ eV the expansion parameter is $k_B T/E_a = 1/30.8$, so the Laplace correction $2k_B T/E_a$ is 6.5 percent; at $\sigma = 0.23$ and $T_{\text{ord}} = 240$ K the

thermodynamic factor is $q(146 \text{ K}) = 0.138$ and $d \ln q/dT = -0.0162 \text{ K}^{-1}$, so the second correction is 7.7 percent with the opposite sign. The two nearly cancel in the value of Θ , which is why the leading form suffices for the barrier, but they do not cancel in its logarithmic derivative, which is what the rate shift below measures, and they are carried separately from here on. The prefactor $k_B T^2/E_a$ has dimensions of temperature, so $v k_B T^2/r E_a$ is dimensionless.

The value of q matters as much as its derivative. Retaining it rather than setting it to unity multiplies Θ by 0.138 at the onset, which at fixed n_0 lowers the barrier the onset returns by $k_B T_{\text{on}} \ln(1/q) = 0.025$ eV and at fixed barrier raises the nucleus density the onset demands by q^{-2} , a factor of 53. No step of the argument may treat the driving force as saturated at 146 K, since the undercooling there is only 94 K against a latent entropy of a fraction of k_B .

Two limits confirm Eq. (S11). As $r \rightarrow 0$ the accumulated radius diverges and $\phi \rightarrow 1$ at every temperature at which v is nonzero, which is the statement that an infinitely slow sweep is an equilibrium measurement. As $r \rightarrow \infty$ the radius vanishes and the sample never transforms, which is the statement that an infinitely fast sweep quenches.

C. The onset and the barrier

Define the onset as the rise midpoint, $\phi(T_{\text{on}}) = 1/2$, which by Eq. (S7) requires

$$\pi n_0 R_w(T_{\text{on}})^2 = \ln 2, \quad R_w(T_{\text{on}}) = \sqrt{\frac{\ln 2}{\pi n_0}}. \quad (\text{S12})$$

The onset therefore fixes not a barrier but a pair, since the same radius can be reached with a larger barrier and a denser seeding or with a smaller barrier and a sparser one; at $n_0 = 10^{-2} \text{ nm}^{-2}$, one nucleus per hundred square nanometres, the radius the sweep must deliver is 4.70 nm. Inserting Eq. (S11) at leading order, taking logarithms and solving for E_a gives Eq. (5) of the main text,

$$E_a = k_B T_{\text{on}} \ln \left[\frac{v a k_B T_{\text{on}}^2 q(T_{\text{on}})}{E_a r R_w(T_{\text{on}})} \right]. \quad (\text{S13})$$

The argument of the logarithm is dimensionless: v carries s^{-1} , a and R_w carry length, $k_B T_{\text{on}}^2/E_a$ carries K, q is dimensionless and r carries K s^{-1} . Equation (S13) is implicit in E_a but converges under direct iteration because E_a appears inside a logarithm. Starting from $E_a = 0.40$ eV with $T_{\text{on}} = 146$ K, $v = 10^{13} \text{ s}^{-1}$, $r = 1 \text{ K/min}$ and $n_0 = 10^{-2} \text{ nm}^{-2}$ the successive iterates are 0.3869, 0.3873, 0.3873 eV, so three passes suffice, and numerical integration of Eq. (S8) without any expansion returns 0.3874 eV, the agreement to one part in four thousand being what remains of the two corrections in Eq. (S11) once they have cancelled against one another.

The error budget follows from differentiating Eq. (S13). Writing $\varepsilon = E_a$, $a_0 = k_B T_{\text{on}}$ and $C = v a k_B T_{\text{on}}^2 q(T_{\text{on}})/(r R_w)$,

so that $\varepsilon = a_0 \ln(C/\varepsilon)$,

$$\frac{\partial \varepsilon}{\partial \ln C} = \frac{a_0 \varepsilon}{\varepsilon + a_0}, \quad \frac{\partial \varepsilon}{\partial T_{\text{on}}} = \frac{\varepsilon/T_{\text{on}} + 2k_B + k_B T_{\text{on}} d \ln q/dT}{1 + a_0/\varepsilon}. \quad (\text{S14})$$

With $\varepsilon = 0.387$ eV and $a_0 = k_B T_{\text{on}} = 12.58$ meV, the first derivative is 12.19 meV per e -fold of C , that is 28 meV per decade; since $C \propto \nu$ and $C \propto 1/r$, a decade of attempt frequency and a decade of sweep rate each move E_a by 0.028 eV, in opposite directions, and since $C \propto R_w^{-1} \propto n_0^{1/2}$, a decade of nucleus density moves it by half as much, 0.014 eV. The second derivative is 2.54 meV/K, whose reciprocal is the 393 K/eV that the width analysis below uses, so the ± 7 K uncertainty on the onset contributes ± 0.018 eV, and carrying the latent entropy over 0.10 to 0.69 k_B per cobalt through $q(T_{\text{on}})$ contributes -0.010 to $+0.012$ eV. Added in quadrature these give 0.047 eV, quoted as $E_a = 0.39 \pm 0.05$ eV. Solving Eq. (S13) at the corners of the plausible ranges $\nu \in [10^{11}, 10^{14}]$ s $^{-1}$, $r \in [0.5, 5]$ K/min and $n_0 \in [10^{-4}, 10^{-1}]$ nm $^{-2}$ gives 0.28 eV at the low corner and 0.44 eV at the high corner, widening to the interval 0.27 to 0.46 eV once the onset and latent-entropy ranges are carried to their corners as well.

The degeneracy between the barrier and the nucleus density is exact enough to be written down. Holding T_{on} at 146 K and everything else at the values above, $E_a = 0.387 + 0.014 \log_{10}(n_0/10^{-2} \text{ nm}^{-2})$ eV, so the plausible band $n_0 = 10^{-4}$ to 10^{-1} nm $^{-2}$, corresponding to nucleus spacings of 100 down to 3 nm, maps onto $E_a = 0.36$ to 0.40 eV. Neither number is measured on its own; what the onset measures is the combination, and the rate series of the next subsection is what separates the barrier from the seeding.

D. The rate shift and its independence of the Avrami exponent

Take the logarithm of the onset condition Eq. (S12), writing $R_w = a\Theta$ and using Eq. (S11) at leading order,

$$\ln(\pi n_0 a^2) + 2 \left[\ln \frac{\nu}{r} + \ln \frac{k_B T_{\text{on}}^2 q(T_{\text{on}})}{E_a} - \frac{E_a}{k_B T_{\text{on}}} \right] = \ln \ln 2. \quad (\text{S15})$$

Differentiate Eq. (S15) with respect to $\ln r$ at fixed E_a , ν and n_0 . The factor of two divides out; inside the bracket the rate term gives -1 , the prefactor gives $2 dT_{\text{on}}/(T_{\text{on}} d \ln r)$, the thermodynamic factor gives $(d \ln q/dT) dT_{\text{on}}/d \ln r$ and the exponential gives $+E_a dT_{\text{on}}/(k_B T_{\text{on}}^2 d \ln r)$, so

$$\frac{dT_{\text{on}}}{d \ln r} = \left[\frac{2}{T_{\text{on}}} + \frac{E_a}{k_B T_{\text{on}}^2} + \frac{d \ln q}{dT} \right]^{-1}, \quad (\text{S16})$$

which is Eq. (6) of the main text and reduces, when the driving force is treated as saturated and $d \ln q/dT$ is dropped, to the Kissinger relation $k_B T_{\text{on}}^2/(E_a + 2k_B T_{\text{on}})$ for a first-order rate process [3]. Numerically, at $T_{\text{on}} = 146$ K the three terms in the bracket are 0.0137, 0.2107 and -0.0162 K $^{-1}$, giving 4.80 K per e -fold, or 11.1 K per decade of warming rate; the thermodynamic factor is what carries the last of the three, and dropping it would return 10.3 K per decade. Across the barrier range that the onset admits at $T_{\text{on}} = 146$ K the same expression

gives 11.6, 11.1 and 10.7 K per decade at $E_a = 0.37, 0.387$ and 0.40 eV, and integrating Eq. (S8) numerically over a decade of rate from 0.5 to 5 K/min gives 11.4 K. The prediction quoted in the main text is therefore 11 ± 1 K per decade, the spread being the width of the admissible barrier range rather than a fitted uncertainty.

Suppose now that the transformed fraction is $\phi = 1 - \exp(-A_0 \Theta^n)$ for a constant A_0 and exponent n , which covers continuous nucleation with growth ($n = 3$ in two dimensions) and growth from a frozen nucleus population ($n = 2$, with $A_0 = \pi n_0 a^2$), as well as the simple relaxation $n = 1$. The onset condition becomes $A_0 \Theta^n = \ln 2$, whose logarithm is

$$n \left[\ln \frac{\nu}{r} + \ln \frac{k_B T_{\text{on}}^2 q(T_{\text{on}})}{E_a} - \frac{E_a}{k_B T_{\text{on}}} \right] = \ln \frac{\ln 2}{A_0}. \quad (\text{S17})$$

Differentiating with respect to $\ln r$ divides out the overall factor n and returns Eq. (S16) unchanged, the constant A_0 dropping out entirely; direct numerical solution of Eq. (S7) at $n = 1$ and $n = 2$ confirms the cancellation to four figures. The onset shift per decade of warming rate is therefore a property of the barrier and the onset temperature alone, and it does not test which transformation mechanism operates. The exponent instead controls the width of the rise, which is what the next subsection quantifies, and it shifts the extracted E_a only through the constant A_0 inside a logarithm, that is by the 0.014 eV per decade of n_0 already budgeted.

The cancellation carries one condition, and the rate series has to be run so as to respect it. Equation (S17) treats A_0 , and with it n_0 , as a constant of the protocol, which it is only if the cooldown that seeds the sample is held fixed while the warming rate is varied. If the nuclei are thermally seeded then their density falls as the cooldown is hurried, $n_0 \propto 1/|r_c|$, and differentiating Eq. (S15) with respect to $\ln |r_c|$ instead gives half the shift of Eq. (S16), that is $+5.5$ K per decade of cooling rate, with the same sign, since a sparser seeding forces the sweep to grow larger domains before the midpoint is reached. A protocol that slaves the cooldown to the warming rate therefore returns about 16.5 K per decade, half as much again as the growth shift alone, and the two contributions cannot be separated after the fact. The measurement that tests Eq. (S16) is a warming series taken at a single fixed cooling rate.

E. Width of the rise

The bracket of Eq. (S16) is $d \ln \Theta/dT$, so the extended area $A_{\text{ex}} = \pi n_0 a^2 \Theta^2$ that sits in the exponent of Eq. (S7) grows by a factor of ten every

$$\delta T_{10\times} = \frac{\ln 10}{2} \left(\frac{d \ln \Theta}{dT} \right)^{-1} = 5.5 \text{ K} \quad (\text{S18})$$

at $T_{\text{on}} = 146$ K and $E_a = 0.387$ eV, half the 11.1 K that Θ itself takes. The transformed fraction passes from 10 to 90 percent as A_{ex} runs from $-\ln 0.9 = 0.105$ to $\ln 10 = 2.303$, a factor of 21.9, so

$$\Delta T_{10-90} = \delta T_{10\times} \log_{10}(21.9) = 7.4 \text{ K}, \quad (\text{S19})$$

which becomes 7.3 K when Eq. (S8) is integrated across the rise rather than expanded about the midpoint, and that is the number carried forward; the corresponding width for a general exponent is larger by the factor $2/n$, so continuous nucleation would be narrower still and simple relaxation twice as broad. The observed width, 19_{-9}^{+6} K, exceeds all of them. A spread of barriers δE_a broadens the onset by $\delta T_{\text{on}} = \delta E_a (\partial E_a / \partial T_{\text{on}})^{-1}$, the second factor being the 393 K/eV that Eq. (S14) returns, so $\delta E_a = 0.045$ eV contributes 17.5 K, which added in quadrature to Eq. (S19) gives 19.0 K. A barrier spread of that size is not an adjustable quantity here: the tetrahedral-site divacancy barrier computed for Li_xCoO_2 scatters by ± 0.04 eV across the local lithium environments of a single composition [4], so the width of the rise and the environment dependence of the barrier are one statement rather than two. The width is accordingly consistent with growth from a frozen nucleus population carrying the environment scatter the first-principles barriers already predict, and is not consistent with a sharply cooperative transformation.

S3. NUCLEATION, GROWTH, AND THE BRANCH ASYMMETRY

A. The two-dimensional nucleation barrier

A circular domain of ordered phase of radius R within the lithium plane costs line tension and gains bulk free energy [5, 6],

$$\Delta G(R) = -\pi R^2 |\Delta f_A| + 2\pi R \lambda, \quad (\text{S20})$$

with Δf_A the free-energy difference per unit area, negative below T_{ord} , and λ the domain-wall line tension, of dimension energy per length. Setting $d\Delta G/dR = 0$ gives the critical radius and the barrier,

$$R^* = \frac{\lambda}{|\Delta f_A|}, \quad W^* = \Delta G(R^*) = \frac{\pi \lambda^2}{|\Delta f_A|}, \quad (\text{S21})$$

whose dimensions are (energy/length)² divided by (energy/area), that is energy. Close to the boundary the driving force is linear in the undercooling, $|\Delta f_A| \simeq \Delta S_A (T_{\text{ord}} - T)$ with ΔS_A the latent entropy per unit area, so

$$\frac{W^*}{k_B T} \simeq \frac{\pi \lambda^2}{k_B \Delta S_A} \frac{1}{T (T_{\text{ord}} - T)}, \quad (\text{S22})$$

which diverges as $T \rightarrow T_{\text{ord}}^-$ and falls monotonically as the undercooling grows.

B. Transformed fraction

With nucleation rate per unit area I_n and growth velocity v as in Eq. (7) of the main text, the extended area fraction of the Kolmogorov–Johnson–Mehl–Avrami construction [7–10] is

$$A_{\text{ex}}(t) = \pi \int_0^t I_n(t') \left[\int_{t'}^t v(t'') dt'' \right]^2 dt', \quad (\text{S23})$$

in which each nucleus born at t' contributes the area of the disk it would have grown by time t had it met no other domain, and the transformed fraction is

$$\phi(t) = 1 - \exp[-A_{\text{ex}}(t)]. \quad (\text{S24})$$

The dimensions check: I_n carries (area time)⁻¹, the bracket carries length, and the outer dt' carries time, so A_{ex} is dimensionless. For constant I_n and v , Eq. (S23) reduces to $A_{\text{ex}} = \pi I_n v^2 t^3 / 3$, the familiar two-dimensional Avrami exponent of three. If instead nucleation is complete before growth begins, with a frozen areal density n_0 of nuclei, then

$$A_{\text{ex}}(t) = \pi n_0 \left[\int_0^t v dt' \right]^2 = \pi n_0 R_w^2, \quad (\text{S25})$$

which is Eq. (S7) and the $n = 2$ member of the $A_0 \Theta^n$ family analyzed in Sec. S2, with $A_0 = \pi n_0 a^2$; the onset shift is then Eq. (S16) and only the width changes.

Which of the two limits applies is decided by the nucleation rate, and the point that governs the whole of this section is that the nucleation rate carries the same mobility factor as the growth velocity,

$$I_n(T) = I_0 e^{-[E_a + W^*(T)]/k_B T}, \quad (\text{S26})$$

with $I_0 = v/a^2$ for homogeneous nucleation on the lithium sublattice and $I_0 = Xv$ for heterogeneous nucleation on a template of areal density X . Only W^* distinguishes Eq. (S26) from Eq. (S6); the ionic mobility that has to unfreeze before a domain can grow has equally to unfreeze before a domain can be born.

C. Why cooling and warming differ

The two exponentials in Eqs. (S6) and (S26) move in opposite directions. By Eq. (S22) the barrier W^* is prohibitive just below T_{ord} and falls monotonically as the undercooling grows, while the mobility factor common to both rates collapses as the temperature falls. Their product has a single maximum, the nose of the time-temperature-transformation curve [11], which for every line tension considered here sits between 178 and 211 K. What it does not have is a second maximum at large undercooling: the product $T(T_{\text{ord}} - T)$ in the denominator of Eq. (S22) is itself largest at $T_{\text{ord}}/2 = 120$ K, so below that temperature $W^*/k_B T$ rises again as the sample cools and joins the mobility factor rather than opposing it, $dI_n/dT > 0$ holds throughout, and the bottom of the sweep is where nucleation is least likely rather than most. At 80 K the mobility factor alone is e^{-56} , and a nucleation event there is as forbidden as a growth event.

The requirement that the smooth cooling trace places on the seeding can therefore be stated as a birth temperature. Smoothness demands that the extended area accumulated on the way down stay below some $A_{\text{ex}}^{\text{max}}$, which we take as 10^{-2} , while the warming branch demands $\pi n_0 R_w(T_{\text{on}})^2 = \ln 2$; since both are quadratic in the radius, any nucleus present at base temperature must have grown on the way down to no more

than $(A_{\text{ex}}^{\text{max}}/\ln 2)^{1/2} = 0.12$ of the radius the warming sweep will give it. The growth accumulated from a birth temperature T' downward is the same Laplace integral as Eq. (S11), evaluated at T' instead of T_{on} , so the ratio of the two radii is $e^{-(E_a/k_B)(1/T' - 1/T_{\text{on}})}$ up to algebraic prefactors of order unity, and the condition is

$$\frac{E_a}{k_B} \left(\frac{1}{T'} - \frac{1}{T_{\text{on}}} \right) \geq \frac{1}{2} \ln \frac{\ln 2}{A_{\text{ex}}^{\text{max}}}. \quad (\text{S27})$$

The right-hand side is 2.12, and the left-hand side is steep, so the ceiling it sets is almost independent of the barrier: $T' \leq 136.2, 136.6, 136.9$ and 137.8 K at $E_a = 0.37, 0.387, 0.40$ and 0.45 eV. Every nucleus the sample carries to base temperature must have been born below $T_{\text{on}} - 9$ K. A nucleus born at the nose, by contrast, grows between 0.6 nm and $40 \mu\text{m}$ over the remainder of a 1 K/min cooldown against the 1.5 to 47 nm that Eq. (S12) asks of it, so a single such nucleus per flake transforms the sample on the way down and destroys the branch asymmetry.

Classical nucleation cannot deliver seeds under that ceiling, and the arithmetic says so without appeal. Pushing the maximum of Eq. (S26) below 137 K while W^* retains the $1/(T_{\text{ord}} - T)$ form of Eq. (S22) requires $|d(W^*/k_B T)/dT|$ to exceed $E_a/k_B T^2 = 0.21 \text{ K}^{-1}$ there, hence $W^*/k_B T \approx 100$, and the rate is suppressed by e^{-100} . Put as a constraint on the one free quantity in Eq. (S22), the line tension is overdetermined: keeping the cooling trace smooth requires λ to be at least about 53 to 59 meV/nm over the admissible barriers, while delivering the seeding density that Eq. (S12) needs requires it to be at most 38 to 48 meV/nm. Scanning the whole of the plausible parameter space, E_a from 0.34 to 0.45 eV, ν from 10^{11} to 10^{14} s^{-1} , σ from 0.10 to 0.69 , λ over four decades and the cooling rate from 1 to 60 K/min, the two requirements never meet: the shortfall in nucleus density is five to eight decades at 1 K/min and never falls below 1.9 decades anywhere in the space, and the corner that comes closest does so only by delivering almost no nuclei at all. Heterogeneous nucleation does not rescue this. A potency factor ψ that replaces W^* by ψW^* is exactly equivalent to replacing λ by $\lambda \sqrt{\psi}$, so it moves within the same one-parameter family already scanned, and it does so with a prefactor $X\nu$ smaller than ν/a^2 by some three orders of magnitude, which costs a further one to one and a half decades. In the opposite limit, where the template removes the barrier altogether, the sites activate as soon as the sample crosses T_{ord} , where Eq. (S28) gives a relaxation time of $3 \times 10^{-4} \text{ s}$, and the flake transforms on the way down.

What the argument leaves is an assumption rather than a derived result. The seeds are taken to be athermal: a quenched population of defect-templated embryos, patches of ordered lithium pinned to structural imperfections of the flake, which are present at all temperatures and go supercritical not by a thermal fluctuation but as soon as the shrinking critical radius of Eq. (S21) falls to their own size. That crossing carries no mobility factor and therefore no $e^{-E_a/k_B T}$ suppression, which is precisely the functional form the classical estimate cannot supply. At 137 K the critical radius is $R^* = \lambda/|\Delta f_A| = 0.67, 1.01$ and 1.35 nm for $\lambda = 20, 30$ and 40 meV/nm, that is patches of $21, 47$ and 83 cobalt sites, so an embryo population whose

typical member is of that size activates in the right window. The assumption carries a condition of its own: the embryo-size distribution has to be cut off above about 1.3 nm, because a single 2 nm embryo anywhere in the flake goes supercritical near 200 K and sweeps $10 \mu\text{m}$ before the cooldown ends. A cutoff of that kind is natural for patches templated by a fixed defect population and unnatural for thermal fluctuations of the parent phase, which is the sense in which the assumption is a physical one rather than a fitted one.

The assumption is not free, because it makes two predictions that the thermal alternative does not. If the seeds are athermal then n_0 is a property of the flake and not of the protocol, so T_{on} is independent of the cooling rate; if they are thermally nucleated then $n_0 \propto 1/|r_c|$ and T_{on} moves by $+5.5$ K per decade of cooling rate, by the differentiation of Eq. (S15) carried out in Sec. S2. One device and one afternoon separate the two. Second, the extended area accumulated on the way down scales as $|r_c|^{-3}$ under continuous nucleation, so a cooldown slower than about 0.1 K/min must transform the sample on the way down and erase the warming anomaly altogether, whichever provenance the seeds have.

On the warming branch none of this is in play. The sample begins at maximum undercooling with the nucleus population already present, so Eq. (S25) applies, nucleation is not rate limiting, and the only barrier left is the one that controls growth. That is why the single-barrier description of Eq. (S6) is legitimate on warming and not on cooling, and it is why the barrier extracted from the onset is a migration barrier rather than a sum of migration and nucleation contributions.

D. The isothermal hold

Stopping the sweep is the sharpest test the mechanism admits, but the two branches test different things and only one of them can falsify.

A hold on the cooling branch, between 150 and 170 K, interrogates the nucleation rate, because that is the quantity the smooth cooling trace constrains. What it constrains is a lower bound. Requiring the extended area accumulated during the sweep's passage through a given temperature to stay below 10^{-2} , with the Avrami exponent of three appropriate to continuous nucleation, already places τ_{nose} above 78 min at 170 K, 2.9 h at 160 K and 10 h at 150 K for a 1 K/min cooldown, and above ten to a hundred minutes at 10 K/min. A hold of an hour that produces nothing is therefore consistent with the model as it stands; it raises the lower bound and nothing more, and a nose time of days is not excluded by any equation here. The cooling-branch hold is worth doing, since a hold that does transform the sample fixes τ_{nose} outright, but a null result there cannot refute the account.

The falsifying test is a hold on the warming branch, below the onset, where the nuclei are present by hypothesis and the transformation is growth limited. The half-transformation time is then the radius the onset demands divided by the growth velocity at the hold temperature, $\tau_{1/2}(T) = R_w(T_{\text{on}})/v(T)$, with both factors already fixed by the onset analysis. Over the admissible window this is 1 to 11 s at 170 K, 5 to 49 s at 160 K,

27 s to 4.5 min at 150 K, 3 to 34 min at 140 K and 9 min to 1.9 h at 135 K, so a hold between 135 and 140 K completes in three minutes to two hours, with the central parameters giving 15 to 50 min. That window is short enough to run and long enough to watch, and a hold there that produces nothing is fatal to the mechanism rather than merely uninformative. The prediction has a parameter-free anchor as well: at the onset itself the sweep has just delivered the critical radius, so $\tau_{1/2}(T_{\text{on}}) = (k_B T_{\text{on}}^2 / E_a) / r$, which is 4 to 5 min at 1 K/min independently of n_0 , ν and λ . Anything much longer than that at T_{on} falsifies the model without reference to any of its free parameters.

There is one empirical anchor for these timescales in the sister compound. Isothermal alkali ordering in $\text{Na}_{0.8}\text{CoO}_2$ proceeds under a hold at 200 K with a time constant near 40 min and saturates only after two to three hours, while a quench to any temperature below about 212 K suppresses it entirely [12], which is the same phenomenology at the same order of magnitude, displaced upward in temperature as the stronger alkali-alkali interaction of the sodium system requires.

E. Quasiequilibrium on the decay side

At the upper end of the anomaly the relaxation time of the hop that carries Eq. (S6),

$$\tau(T) = \nu^{-1} e^{E_a/k_B T}, \quad (\text{S28})$$

equals 1.4×10^{-5} s at $T = 240$ K for $E_a = 0.387$ eV and $\nu = 10^{13}$ s $^{-1}$, and 3×10^{-4} s at the top of the admissible barrier range, which is shorter than a full sweep at 1 K/min by more than seven orders of magnitude. The decay of the anomaly is therefore a thermodynamic collapse of ϕ_{eq} together with the shrinking transport contrast, as in Eq. (8) of the main text, and not a second kinetic process.

This also fixes how far the collapse can be dragged by the sweep, which is the prediction entered in the first row of Table II of the main text. A phase boundary crossed at rate r lags its equilibrium position by the distance the temperature moves within one relaxation time,

$$\delta T_{\text{ord}} \simeq \tau(T_{\text{ord}}) r = \nu^{-1} e^{E_a/k_B T_{\text{ord}}} r, \quad (\text{S29})$$

which at $T_{\text{ord}} = 240$ K and the fastest rate contemplated here, $r = 10$ K/min = 0.17 K s $^{-1}$, is 2×10^{-6} K at the central barrier and 5×10^{-5} K at 0.45 eV. The lag is exponentially sensitive to the barrier and to nothing else, so the bound is quoted at the top of the range rather than at the middle of it: even at 0.50 eV, above the interval the onset admits, Eq. (S29) gives 5×10^{-4} K. The collapse is therefore immovable below the 10^{-3} K level across the whole rate series and the whole barrier range, against an onset that moves by 11 K per decade. The contrast between Eqs. (S16) and (S29) is the sharpest single statement the model makes.

S4. FIELD THERMODYNAMICS OF THE COLLAPSE

A. Clausius–Clapeyron relation

Let f_P and f_O be the free energies per cobalt of the two phases in a field B , with differentials $df = -S dT - M dB$, where S and M are the entropy and the magnetization per cobalt. The coexistence line $T_{\text{ord}}(B)$ is defined by

$$\Delta f(T_{\text{ord}}(B), B) = 0, \quad \Delta f \equiv f_O - f_P. \quad (\text{S30})$$

Differentiating Eq. (S30) totally with respect to B ,

$$\frac{\partial \Delta f}{\partial T} \frac{dT_{\text{ord}}}{dB} + \frac{\partial \Delta f}{\partial B} = 0, \quad (\text{S31})$$

and using $\partial \Delta f / \partial T = -(S_O - S_P) = \Delta S$ and $\partial \Delta f / \partial B = -(M_O - M_P) = -\Delta M$, with the definitions $\Delta S \equiv S_P - S_O$ and $\Delta M \equiv M_O - M_P$ of the main text, gives Eq. (9),

$$\frac{dT_{\text{ord}}}{dB} = \frac{\Delta M(B)}{\Delta S}. \quad (\text{S32})$$

The signs are worth stating explicitly. The ordered phase has fewer accessible configurations, so $S_O < S_P$ and $\Delta S > 0$. The measured collapse moves upward in field, so $dT_{\text{ord}}/dB > 0$, and Eq. (S32) then forces $\Delta M > 0$: the ordered phase is the one with the larger magnetization. Dimensionally, magnetization per site divided by entropy per site is (energy/field) divided by (energy/temperature), that is temperature per field, as required.

B. Numerical conversion

Write $\Delta M = m \mu_B$ and $\Delta S = \sigma k_B$ per cobalt. Then

$$\begin{aligned} \frac{dT_{\text{ord}}}{dB} &= \frac{m}{\sigma} \frac{\mu_B}{k_B}, \\ \frac{\mu_B}{k_B} &= \frac{5.788 \times 10^{-5} \text{ eV/T}}{8.617 \times 10^{-5} \text{ eV/K}} = 0.6717 \text{ K/T}. \end{aligned} \quad (\text{S33})$$

The measured secant slope is $\Delta T_{\text{ord}}/\Delta B = (7 \pm 7) \text{ K}/(5 \text{ T}) = 1.4 \pm 1.4 \text{ K/T}$, so

$$m = \frac{1.4 \pm 1.4}{0.6717} \sigma = (2.1 \pm 2.1) \sigma, \quad (\text{S34})$$

which is Eq. (10) of the main text.

Everything here is per cobalt *of the phase that transforms*, and the bound below is quoted for the reading in which that phase is the bulk. At $x \simeq 1/2$ half the bulk cobalt sites are Co^{4+} in a low-spin d^5 configuration carrying $S = 1/2$, so the maximum magnetization the ordered phase can carry is $m = 0.48$ per cobalt with every such moment aligned, and the parent phase, being a Pauli-paramagnetic metal, contributes negligibly. Section S5 turns that ceiling from an estimate into a computed quantity, the ground state returning $0.97 \mu_B$ on the Co^{4+} sites and $0.00 \mu_B$ on the Co^{3+} sites. Equation (S34) then bounds the latent entropy,

$$\sigma \lesssim \frac{0.48}{2.1} \simeq 0.23, \quad (\text{S35})$$

against the $\ln 2 = 0.69$ per cobalt that complete lithium order and disorder at half filling would release, so under the bulk reading the transformation converts at most about a third of the available configurational entropy, which is consistent with the partial ordering that the broadened rise of Sec. S2 also indicates. Equation (S35) is not an independent statement from the headcount derived below: inverting Eq. (S37) for σ gives $\sigma(N) = m_{\text{sat}} \tanh(N\mu_B B/k_B T)/2.1$, which approaches $m_{\text{sat}}/2.1$ as the block grows without limit, so 0.23 is the full-polarization end of the same curve whose other end fixes N . The two are an upper and a lower reading of one relation rather than two independent results. The same calculations show that this ceiling is never approached in the bulk, since the dominant coupling is antiferromagnetic, $E_{\text{AFM}} - E_{\text{FM}} = -56.5$ meV per Co^{4+} within a row, and both computed ground states carry zero net moment, so the magnetization that tilts the collapse line has to come from elsewhere. If it resides in a surface gas, the measured slope refers to surface cobalt and the latent entropy per bulk cobalt is smaller by the surface-to-volume ratio, about 10^{-2} for a flake 20 to 60 nm thick. Section VII of the main text pairs calorimetry with the thickness series to decide between the two readings.

C. Induced moments

If the phases carry no permanent moments and respond only through their susceptibilities, the field term in the free-energy difference is $-\frac{1}{2}\Delta\chi B^2$ rather than $-\Delta MB$, and Eq. (S32) integrates to

$$T_{\text{ord}}(B) - T_{\text{ord}}(0) = \frac{\Delta\chi B^2}{2\Delta S}. \quad (\text{S36})$$

The quantity extracted from a single 5 T datum is then a secant, and the terminal slope at 5 T is twice as large, so the moment demanded at 5 T becomes $m = (4.2 \pm 4.2)\sigma$. The induced-moment reading therefore requires more magnetization, not less, and strengthens rather than weakens the headcount of the next subsection. Equation (S36) also predicts that the collapse shift grows quadratically with field, which the field series of the main text distinguishes from the saturating shift of aligned permanent moments.

D. Moment headcount

The question the headcount answers is one of polarization rather than of energy balance. The free-energy comparison is already exhausted by the Clausius–Clapeyron relation, which fixes the product ΔMB against $\Delta S dT_{\text{ord}}$ and says nothing about how many moments carry the magnetization; what the measurement additionally demands is that the ordered phase actually be magnetized by $m = 2.1\sigma$ in units of μ_B per cobalt at 5 T, and an independent spin cannot deliver it. At $B = 5$ T the single-moment Zeeman energy is $g\mu_B B = 2 \times 5.788 \times 10^{-5} \times 5 = 0.579$ meV against $k_B T$ of 12.93, 15.77 and 20.68 meV at 150, 183 and 240 K, the onset of the ordered phase, the peak of the field-doubled excess, and

TABLE S1. The size of the field-coupled block, Eq. (S38), at $B = 5$ T and $m_{\text{sat}} = 0.48$, against the latent entropy $\sigma = \Delta S/k_B$ per cobalt and the three temperatures at which the field effect is read. The last row is the ceiling $\sigma = m_{\text{sat}}/2.1 = 0.23$ of Eq. (S35), at which the block is fully polarized and N diverges.

σ (k_B/Co)	N at 150 K	N at 183 K	N at 240 K
0.02	3.9	4.8	6.3
0.05	9.9	12.1	15.9
0.08	16.3	19.9	26.1
0.10	21.0	25.6	33.5
0.15	35.1	42.8	56.2
0.20	60.5	73.8	96.8
0.23	∞	∞	∞

the collapse, so a lone spin is outmatched by factors of 22, 27 and 36 and is polarized only to $\tanh(\mu_B B/k_B T) = 0.022$, 0.018 and 0.014. The field-coupled unit must therefore be a block of N moments locked parallel by exchange, whose Zeeman energy is $N\mu_B B$ and whose polarization along the field is correspondingly larger.

Write m_{sat} for the magnetization per cobalt the ordered phase would carry with every Co^{4+} moment aligned, which Sec. S5 computes as $m_{\text{sat}} = 0.48$. A block of N such moments is polarized to

$$m = m_{\text{sat}} \tanh\left(\frac{N\mu_B B}{k_B T}\right), \quad (\text{S37})$$

and setting this equal to the $m = 2.1\sigma$ that Eq. (S34) demands and solving for N gives Eq. (11) of the main text,

$$N = \frac{k_B T}{\mu_B B} \operatorname{arctanh}\left(\frac{2.1\sigma}{m_{\text{sat}}}\right), \quad (\text{S38})$$

which reduces to $N \simeq (2.1\sigma/m_{\text{sat}})(k_B T/\mu_B B)$ whenever the block is far from saturation. The criterion carries no arbitrary constant: the latent entropy σ enters explicitly, so that quoting a headcount requires quoting the entropy it assumes. Table S1 evaluates Eq. (S38) across the entropies the transformation admits and the three temperatures of the window.

At the latent entropy of about a tenth of k_B per cobalt that the calorimetry row of Table II of the main text carries, the block holds twenty-one moments at 150 K and twenty-six at 183 K; at half that entropy it holds ten and twelve. Across the entropies that a partial conversion allows, 0.02 to 0.10 k_B per cobalt, and across the 150 to 183 K window in which the field-doubled excess is read, N runs from about four to about twenty-six, so the field-coupled object holds of order ten aligned moments and the range worth quoting is five to twenty-five. Figure 3(b) of the main text draws Eq. (S37) against N at 150 K together with the magnetization the measured slope demands, and marks the crossings at $N \simeq 10$ for $\sigma = 0.05$ and $N \simeq 21$ for $\sigma = 0.10$, the curve saturating at m_{sat} .

The energy scale required of the field-induced change in transport is a separate calculation that happens to share a logarithm. Doubling an activated conductivity ratio requires an

extra

$$\delta E = k_B T \ln 2 = \begin{cases} 9.0 \text{ meV}, & T = 150 \text{ K}, \\ 10.9 \text{ meV}, & T = 183 \text{ K}, \end{cases} \quad (\text{S39})$$

of transport gap in the ordered phase, which is what Eq. (12) of the main text asks the block of Eq. (S38) to supply. The $\ln 2$ here is the factor of two in a conductivity ratio and nothing else; it is not a criterion for N , and it is unrelated to the $\ln 2$ that defines the rise midpoint in Eq. (S12) or to the $\ln 2$ per cobalt of complete lithium disorder in Eq. (S35). The three appear in three different equations for three different reasons.

S5. DENSITY-FUNCTIONAL COMPUTATIONAL DETAILS

All calculations use QUANTUM ESPRESSO 7.3.1 [13, 14] with the PBE exchange-correlation functional [15] and projector augmented-wave datasets [16] for Co, O and Li, plane-wave cutoffs of 60 Ry for the wave functions and 480 Ry for the charge density, Marzari–Vanderbilt cold smearing [17] of width 0.01 Ry, and a self-consistency threshold of 10^{-8} Ry. The Hubbard term is applied to the Co $3d$ shell in the rotationally invariant simplified form [18] with atomic projectors.

A. Why $U = 4.0$ eV, and why it is not a free parameter

The value is fixed by a physical requirement rather than chosen for agreement. A scan over $U = 0, 2.5, 4.0$ and 5.5 eV gives four qualitatively different materials. At $U = 0$ every cell is an itinerant metal with no site resolution. At $U = 2.5$ eV the moment settles on the lithium-rich cobalt rather than the lithium-poor one, so that the sign of the charge disproportionation is inverted, and the $x = 1/3$ cell is a half-metallic ferromagnet with a noninteger moment. At $U = 5.5$ eV the Co^{4+} site is driven high spin, carrying $2.28 \mu_B$, and the $x = 1/2$ cell is metallic again. Only near $U = 4.0$ eV does the calculation return the low-spin $S = 1/2$ Co^{4+} and $S = 0$ Co^{3+} pair that the chemistry of Li_xCoO_2 requires. The Hubbard parameter is therefore calibrated to a known spin state, and every number reported below carries that calibration with it. A metal-insulator transition sits between $U = 2.5$ and 4.0 eV at $x = 1/3$; above the transition the gap is nearly independent of U , moving from 1.179 to 1.166 eV between 4.0 and 5.5 eV while the moment continues to grow, which is the signature of a gap set by the ordering pattern rather than by U itself.

B. Cells

The row-ordered $x = 1/2$ structure is the one-layer monoclinic cell of the Reimers–Dahn and Van der Ven ground state [19, 20], space group $P2_1/m$, with two Co, four O and one Li at the experimental $a_{\text{hex}} = 2.808 \text{ \AA}$ and $c_{\text{hex}} = 14.24 \text{ \AA}$. The third lattice vector carries the O3 interlayer offset, so a single

CoO_2 layer reproduces the ABC stacking and the cell is a one-layer setting equivalent to the reported monoclinic cell, with $\beta_m = 108.9^\circ$. The two cobalt sites are inequivalent by lithium coordination, two neighbors against four, which permits charge order without imposing it; they are given distinct species labels sharing one pseudopotential and one U , purely so that symmetrization cannot average them. Antiferromagnetic arrangements are treated in two supercells, one doubled along the row direction so that Co^{4+} moments alternate within a row at $2.808 \text{ \AA}$, and one doubled across the rows so that adjacent rows alternate at $4.864 \text{ \AA}$, each carrying its own ferromagnetic and nonmagnetic references so that every comparison is made inside a single cell. Sampling is $12 \times 8 \times 8$ for self-consistency and $16 \times 10 \times 10$ for the density of states in the primitive cell, with equivalent densities in the supercells.

The motif comparison uses a 4×2 in-plane supercell, eight Co, sixteen O and four Li, that is twenty-eight atoms and eight formula units. A 2×2 cell cannot do this job: on a triangular lattice at half filling all six placements of two lithium among four sites are nearest-neighbor pairs generating the same stripe order in one of three equivalent directions, so row, zigzag and clumped are literally the same structure there. In the 4×2 cell the row and zigzag arrangements both achieve the minimum lithium-lithium coordination of two, while the clumped arrangement has four. Sampling is $3 \times 6 \times 4$ for self-consistency and $4 \times 8 \times 4$ for the density of states, equivalent to $12 \times 12 \times 4$ and $16 \times 16 \times 4$ in the primitive cell. In the row arrangement every cobalt sees exactly three lithium, so the lithium potential is uniform over the cobalt sublattice and any charge order there is spontaneous; the $U = 0$ control confirms this, the row cell relaxing to a uniform cobalt sublattice with moments of 0.3990 and $0.3986 \mu_B$.

The $x = 1/3$ calculation uses the $\sqrt{3} \times \sqrt{3} R30^\circ$ ordering with three Co, six O and one Li per cell, sampled at $7 \times 7 \times 6$ and $12 \times 12 \times 8$.

C. Caveats

The $x = 1/2$ cells are built at $c_{\text{hex}} = 14.24 \text{ \AA}$, which rounds the $14.2235 \text{ \AA}$ interlayer spacing reported for $\text{Li}_{0.68}\text{CoO}_2$ [21] rather than the value at the half-lithiated composition, where the measurements give 14.42 to $14.44 \text{ \AA}$ [19, 22]. The calculations therefore sit at an interlayer distance compressed by about one percent relative to the composition they describe, and since both the tetrahedral-site migration barrier and the oxygen-mediated interlayer coupling are known to respond to the c parameter, this is a geometry limitation that attaches to every number computed in the row-ordered and motif cells. No number reported here has been rescaled to compensate for it, and the comparisons that carry the argument are made between cells sharing the same geometry, so the residual enters as a common offset rather than as a contrast.

Hubbard projectors are atomic rather than orthogonalized atomic. Orthogonalized projectors were tried first and were unaffordable, a single force evaluation on the fourteen-atom cell running beyond twenty-three minutes because the derivative of the orthogonalization matrix is not accelerated in this release,

and on the ten-atom cell they also prevented self-consistency from converging because two near-degenerate t_{2g} Hubbard eigenvalues rotate into each other between iterations. Every comparison reported here uses atomic projectors throughout, so the choice is internally consistent, but absolute occupation traces are projector-dependent and the charge-order claim rests on the differences between cobalt sites, which are corroborated independently by Löwdin d occupancies and by atomic-sphere charges.

The ionic relaxation of the $x = 1/2$ cell was stopped on a force plateau. The total energy became stationary to 0.27 meV per cell over the last six steps while the reported force settled at 78 meV/Å rather than reaching the 25 meV/Å target. The residual sits entirely on the oxygen z coordinates in exact inversion pairs with the energy flat, so it is a systematic force-expression residual rather than a real displacement, and displacements consistent with it change the energy by less than 0.1 meV, far below the magnetic differences of tens of meV reported here. Cobalt and lithium sit on inversion centers and carry identically zero force.

The motif comparison is made at fixed ideal geometry, identical for the three arrangements, so that the comparison is internally consistent; the residual forces are nearly the same in all three and are dominated by the common Co-O breathing that accompanies charge order rather than by anything specific to the lithium configuration, so relaxation should shift the three energies together. Ferromagnetic alignment is imposed in that campaign, and antiferromagnetic alignments of the Co^{4+} moments were not explored there.

Both the motif cells and the $x = 1/3$ cell carry one CoO_2 layer. In the motif cells the third lattice vector is the true O3 rhombohedral translation, so those are bulk O3 crystals rather than slabs, but the lithium pattern is forced to repeat identically in every layer and the specific monoclinic interlayer registry of real $\text{Li}_{0.5}\text{CoO}_2$ is not sampled. The $x = 1/3$ cell has the exact interlayer spacing but O1 rather than O3 registry; the gap of 1.2 eV is far larger than any plausible interlayer bandwidth and the exchange splitting is an in-plane quantity, so neither conclusion is at risk.

Runs seeded from the high-spin start prescribed at the outset, in which `starting_magnetization` = 0.3 places $5.1 \mu_B$ on every cobalt because this cobalt dataset carries seventeen valence electrons, mostly failed to converge: seven of twenty-three runs were abandoned after sixty to a hundred and sixteen iterations with residuals stuck between 10^{-2} and 10^{-4} Ry, and are excluded. Where such a run stabilized enough to be read, it sat about 1.35 eV per cobalt above the converged low-spin state, so the low-spin manifold is the right one by a wide margin, but the high-spin branch was not exhaustively excluded at every U .

One methodological warning is worth recording on its own. In the seven-atom primitive cell the ferromagnetic state comes out 3.1 meV per cobalt *above* the nonmagnetic one at $U = 4.0$ eV, which taken alone would say the ordered phase is nonmagnetic. Both supercells say the opposite, by 26 to 36 meV per cobalt, and the nonmagnetic anchor itself agrees across all three cells to better than 0.1 meV per cobalt. What differs is how much symmetry the magnetic solution is

permitted to break: four symmetry operations survive in the primitive cell against two in the supercells, and the supercell solution indeed carries inequivalent Co^{4+} moments of 1.028 and $0.952 \mu_B$, which the primitive cell cannot express. The same primitive cell puts the magnetic state below the nonmagnetic one at every other U tested. A direct control with symmetry switched off did not converge and is reported as inconclusive, so the conclusion rests on the agreement of the two supercells and on the cell independence of the nonmagnetic anchor. The practical lesson holds regardless: a single primitive-cell calculation would have returned the wrong sign for the magnetic instability in this material.

Finally, the disproportionation is about $0.2 e$ rather than the $1 e$ of a formal-valence picture, the usual situation in DFT+ U charge-ordered oxides where the formal difference is screened by oxygen rehybridization. The labels Co^{3+} and Co^{4+} describe the moment cleanly and the charge only qualitatively. Gaps at $U = 4.0$ eV are larger than experiment suggests for $\text{Li}_{0.5}\text{CoO}_2$ and should not be quoted as predictions of a gap magnitude; what survives the choice of U is that the magnetic charge-ordered solutions are gapped while the nonmagnetic ones are metallic, and that the three lithium motifs are indistinguishable, the latter holding at $U = 0$ as well.

S6. PREDICTIONS AND DISCRIMINATING MEASUREMENTS

Table S2 collects the out-of-sample predictions that Sec. VII of the main text draws from the model, each set beside the measurement that decides it and with the numbers and the tolerances that the derivations above supply. Every entry is out of sample in the sense that no number in it was used in fixing the four parameters of the model, and every entry can be performed on existing devices with existing instrumentation.

S7. SYMBOLS

Table S3 collects every symbol that carries physical content in the main text and in this Supplemental Material, together with its dimension, so that a symbol appearing in a caption and the same symbol appearing inside a derivation can be checked against one another.

TABLE S2. Out-of-sample predictions of the model and what each measurement decides. Rates are warming rates; fields are applied perpendicular to the flake unless stated.

Measurement	Prediction	What it decides
Warming-rate series, 0.2 to 10 K/min, cooling protocol held fixed	Onset moves by 11 ± 1 K per decade, Eq. (6) of the main text with the thermodynamic factor retained, steepening toward 16 K per decade if the cooldown is slaved to the warm-up and the seeds are thermally born; the collapse does not move, its residual kinetic lag $\tau(T_{\text{ord}})r$ of Eq. (S29) being no larger than 5×10^{-5} K at 10 K/min and 5×10^{-6} K at 1 K/min anywhere in the admitted barrier range, and below 10^{-3} K even at 0.50 eV	A flat onset kills kinetic unfreezing; a collapse that follows the onset kills the thermodynamic assignment of T_{ord} . The alkali order-disorder transition of the sister compound is sluggish enough to have been followed at more than one sweep rate [23]
Isothermal hold at 135 to 140 K on the warming branch	Growth from seeds already present completes in three minutes to two hours, centrally fifteen to fifty minutes, after which the remaining rise is absent. A hold at 150 to 170 K on the way down is a different measurement, bounded only from below at one to ten hours at 1 K/min. The measured anchor in the sister compound is a time constant of 40 min at 200 K, with ordering suppressed entirely by a quench below 212 K [12]	A null result on the warming branch kills the growth-from-seeds account of Sec. IV C of the main text; a null result on the cooling branch excludes nothing
Cooling-rate series, 0.05 to 10 K/min, warming rate held fixed	Athermal, defect-templated seeds leave the onset where it is; thermally born seeds accumulate as $1/ r_{\text{cool}} $ and raise the onset by 5.5 K per decade of cooling rate. A cooldown slower than about 0.1 K/min transforms the flake on the way down and erases the warming anomaly	Settles the provenance of the seeds, which is the one ingredient of Sec. IV C of the main text assumed rather than derived
Field series at 1, 2, 3, 5, 7, 9 T, fixed protocol	Saturating Brillouin growth for aligned permanent moments, B^2 growth for an induced moment, threshold behavior for spin-dependent percolation	Separates the three field couplings with no adjustable parameter
In-plane against perpendicular field	Nearly isotropic response for spin coupling, $\cos \theta$ for orbital coupling	Selects spin against orbital coupling
Thickness series at fixed batch	Excess sheet conductance proportional to thickness for a volume phase, thickness-independent for a surface phase	Decides volume against surface, and thereby whether the surface electron gas is the ordered phase
Channel-length scaling, four-probe measurement, and a bias-history series	ΔR proportional to channel length for a channel origin and independent of it for a contact origin; amplitude independent of read voltage and of dwell time at bias if no lithium space charge accumulates at the electrodes	Confirms or kills the contact-artifact alternative of Sec. II of the main text, including the Debye-thin space-charge layer that the blocking-contact argument leaves standing
Batch series in x at fixed protocol	T_{on} moves only through the composition dependence of E_a , at $dT_{\text{on}}/dE_a = 393$ K/eV, so the ± 0.04 eV environment scatter of the divacancy barrier [4] allows at most about ± 16 K across $x = 0.37$ to 0.48 , and the same δE_a must appear in the width of the rise; T_{ord} may move by tens of kelvin	An onset shift beyond about 20 K, or one uncorrelated with the width, implicates inhomogeneity or a second transition rather than a single kinetic edge
Calorimetry across the warming anomaly, read together with the thickness series	A bulk transformation gives a latent entropy of order $0.1 k_B$ per Co, that is a latent heat of 2.1 meV per Co, 0.20 kJ/mol or 2.1 J/g, some twenty times a routine sensitivity of 0.1 J/g and comparable with the 82 and 272 J/mol measured on bulk crystals at $x = 0.50$ and 0.67 [22], together with a relative excess independent of thickness; a surface one gives 0.02 J/g, five times below sensitivity, together with a relative excess inversely proportional to thickness	The two measurements together fix which cobalt population Eq. (10) of the main text refers to, and convert it from a ratio into a measurement of ΔM . No ordering exotherm on warming has been reported in any alkali cobaltate, so this is a new measurement rather than a confirmation
Diffraction or phonon spectroscopy across the anomaly, set against tunneling spectroscopy	Three separable observables. The interlayer spacing and the $(00l)$ positions do not move. The in-plane metric changes by at most a tenth of a percent, with a shear below one degree, and the lithium-contrast superlattice reflections are weak enough to require electron diffraction. The site-mean Co-O bond lengths of the two cobalt sites are computed to differ by only 0.001 to 0.003 Å, a magnitude that is strongly U -sensitive and is not itself a prediction of the method, while a hard gap opens in the local spectrum	Separates electronic charge order from a displacive or Jahn-Teller transition, which would contrast the two sites by tens of picometers, and from a stacking glide, which would collapse the interlayer spacing by six percent
Electron or synchrotron diffraction on one flake at 100 K and at 300 K	Weak lithium-contrast superlattice intensity present in the warmed-through low-temperature state and absent in the as-cooled state, with the interlayer spacing and the $(00l)$ positions unchanged to better than half a percent throughout	Confirms that what orders is the lithium sublattice, and excludes the O3 to H1-3 stacking glide of Sec. II A of the main text, which collapses the interlayer spacing by about six percent

TABLE S3. The symbols that carry physical content in the main text and in this Supplemental Material, with their dimensions. The left panel collects the transport and rate quantities, the right panel the nucleation, thermodynamic and electronic-structure quantities. Purely local auxiliaries introduced and discharged inside a single derivation, such as the integration variable y of Eq. (S9) and the substitutions ε , a_0 and C of Eq. (S14), are defined where they appear and are not repeated here. No symbol in the list below carries a second meaning anywhere in either document.

Symbol	Meaning	Dimension	Symbol	Meaning	Dimension
ϕ	transformed volume fraction	dimensionless	I_n, I_0	nucleation rate per area and its prefactor	$\text{m}^{-2}\text{s}^{-1}$
ϕ_{eq}	equilibrium transformed fraction	dimensionless	v, v_0	domain growth velocity and its prefactor	m s^{-1}
x	lithium content of Li_xCoO_2	dimensionless	n_0	frozen nucleus areal density	m^{-2}
q	thermodynamic factor $1 - e^{\Delta f/k_B T}$	dimensionless	a	lithium-lithium spacing in the plane	m
s	resistivity ratio ρ_0/ρ_p	dimensionless	λ	domain-wall line tension	eV m^{-1}
c	mixture geometry factor	dimensionless	W^*, R^*	nucleation barrier and critical radius	eV, m
R_{meas}	measured two-probe resistance	Ω	A_{ex}	extended area fraction	dimensionless
R_p	channel resistance of the parent phase	Ω	A_0, n	Avrami prefactor and exponent	dimensionless
R_c, R_{ch}	contact and channel resistance	Ω	Δf	free-energy difference per cobalt	eV
ΔR	warming-minus-cooling excess	Ω	Δf_A	free-energy difference per unit area	eV m^{-2}
L, A	channel length and cross section	m, m^2	ΔS	latent entropy per cobalt	eV K^{-1}
$k(T)$	rearrangement rate	s^{-1}	ΔS_A	latent entropy per unit area	$\text{eV K}^{-1}\text{m}^{-2}$
ν	attempt frequency	s^{-1}	σ	ΔS in units of k_B	dimensionless
E_a	effective migration barrier	eV	ΔM	magnetization difference per cobalt	eV T^{-1}
δE_a	spread of barriers across the sample	eV	m	ΔM in units of μ_B	dimensionless
r	sweep rate	K s^{-1}	$\Delta\chi$	susceptibility difference per cobalt	eV T^{-2}
T_0	temperature at which the sweep starts	K	B	applied magnetic field	T
Θ	accumulated attempts on the sweep	dimensionless	g	spectroscopic splitting factor	dimensionless
$\delta T_{10\times}$	temperature per decade of A_{ex}	K	N	moments in the field-coupled block	dimensionless
ΔT_{10-90}	10 to 90 percent rise width	K	δE	field-induced change of transport gap	eV
R_w	growth radius accumulated on warming	m	m_{sat}	ΔM with every Co^{4+} moment aligned	dimensionless
T_{on}	rise midpoint of the anomaly	K	U	Hubbard parameter on Co $3d$	eV
T_{ord}	first-order collapse temperature	K	$D(E_F)$	density of states at the Fermi level	eV^{-1}
δT_{ord}	kinetic lag of the collapse	K	n_d	cobalt $3d$ occupation	dimensionless
τ	relaxation time $\nu^{-1} e^{E_a/k_B T}$	s	β_m	monoclinic cell angle	degrees
τ_{nose}	transformation time at the nose	s	$\mathcal{H}$	thermal history of the sample	dimensionless
$\tau_{1/2}$	isothermal half-transformation time	s	X	areal density of nucleation templates	m^{-2}
ψ	heterogeneous nucleation potency	dimensionless			

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